PROJECT FILE

WORKING WITH MONGO DB DOCUMENTS AND COLLECTIONS

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CREATE OPERATION- The **Create** operation is used to add new documents to a collection. MongoDB provides methods like insertOne() for single documents and insertMany() for multiple documents.

Read Operation- The Read operation retrieves documents from a collection. MongoDB uses methods like find() to fetch multiple documents and findOne() to retrieve a single document.

```
> use("CRUD")
< switched to db CRUD
> db["CRUD"].find()
<
> db.CRUD.insertOne({rollno:1,name:"Sam",age:20,department:ECA})
✖ ▸ ReferenceError: ECA is not defined
> db.CRUD.insertOne({rollno:1,name:"Sam",age:20,department:"ECA"})
< {
    acknowledged: true,
    insertedId: ObjectId('6aae544e05178ee015c9caf2')
  }
> db.CRUD.find()
< {
    _id: ObjectId('6aae544e05178ee015c9caf2'),
    rollno: 1,
    name: 'Sam',
    age: 20,
    department: 'ECA'
  }
> db.CRUD.insertMany([{rollno:2,name:"Ram",age:20,department:"ECA"},{roll:3,name:"Adi",age:20,department:"ECA"},{roll:4,name:"E
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6aae5e6505178ee015c9caf3'),
      '1': ObjectId('6aae5e6505178ee015c9caf4'),
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6aae5e6505178ee015c9caf3'),
      '1': ObjectId('6aae5e6505178ee015c9caf4'),
      '2': ObjectId('6aae5e6505178ee015c9caf5'),
      '3': ObjectId('6aae5e6505178ee015c9caf6'),
      '4': ObjectId('6aae5e6505178ee015c9caf7'),
      '5': ObjectId('6aae5e6505178ee015c9caf8'),
      '6': ObjectId('6aae5e6505178ee015c9caf9'),
      '7': ObjectId('6aae5e6505178ee015c9cafa'),
      '8': ObjectId('6aae5e6505178ee015c9cafb'),
      '9': ObjectId('6aae5e6505178ee015c9cafc'),
      '10': ObjectId('6aae5e6505178ee015c9cafd'),
      '11': ObjectId('6aae5e6505178ee015c9cafe'),
      '12': ObjectId('6aae5e6505178ee015c9caff'),
      '13': ObjectId('6aae5e6505178ee015c9cb00'),
      '14': ObjectId('6aae5e6505178ee015c9cb01'),
      '15': ObjectId('6aae5e6505178ee015c9cb02'),
      '16': ObjectId('6aae5e6505178ee015c9cb03'),
      '17': ObjectId('6aae5e6505178ee015c9cb04'),
      '18': ObjectId('6aae5e6505178ee015c9cb05'),
      '19': ObjectId('6aae5e6505178ee015c9cb06'),
      '20': ObjectId('6aae5e6505178ee015c9cb07'),
      '21': ObjectId('6aae5e6505178ee015c9cb08'),
      '22': ObjectId('6aae5e6505178ee015c9cb09'),
      '23': ObjectId('6aae5e6505178ee015c9cb0a'),
      '24': ObjectId('6aae5e6505178ee015c9cb0b'),
      '25': ObjectId('6aae5e6505178ee015c9cb0c'),
      '26': ObjectId('6aae5e6505178ee015c9cb0d'),
      '27': ObjectId('6aae5e6505178ee015c9cb0e'),
      '28': ObjectId('6aae5e6505178ee015c9cb0f'),
      '29': ObjectId('6aae5e6505178ee015c9cb10'),
      '30': ObjectId('6aae5e6505178ee015c9cb11'),
      '31': ObjectId('6aae5e6505178ee015c9cb12'),
      '32': ObjectId('6aae5e6505178ee015c9cb13'),
      '33': ObjectId('6aae5e6505178ee015c9cb14'),
      '34': ObjectId('6aae5e6505178ee015c9cb15'),
      '35': ObjectId('6aae5e6505178ee015c9cb16'),
      '36': ObjectId('6aae5e6505178ee015c9cb17'),
      '37': ObjectId('6aae5e6505178ee015c9cb18'),
      '38': ObjectId('6aae5e6505178ee015c9cb19'),
      '39': ObjectId('6aae5e6505178ee015c9cb1a')
    }
  }
```

```
> db.CRUD.find()
< {
    _id: ObjectId('6aae544e85178ee815c9caf2'),
    rollno: 1,
    name: 'Sam',
    age: 20,
    department: 'ECA'
  }
  {
    _id: ObjectId('6aae5e6585178ee815c9caf3'),
    rollno: 2,
    name: 'Ram',
    age: 20,
    department: 'ECA'
  }
  {
    _id: ObjectId('6aae5e6585178ee815c9caf4'),
    roll: 3,
    name: 'Adi',
    age: 20,
    department: 'ECA'
  }
  {
    _id: ObjectId('6aae5e6585178ee815c9caf5'),
    roll: 4,
    name: 'Era',
    age: 20,
    department: 'ME'
  }
  {
    _id: ObjectId('6aae5e6585178ee815c9caf6'),
    roll: 5,
    name: 'Bee',
    age: 20,
    department: 'CSE'
  }
  {
    _id: ObjectId('6aae5e6585178ee815c9caf7'),
    roll: 6,
    name: 'Arif',
    age: 21,
    department: 'CSE'
  }
{
  _id: ObjectId('6aae5e6585178ee815c9cafe'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caff'),
  roll: 14,
  name: 'Mukesh',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb00'),
  roll: 15,
  name: 'Raheem',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb01'),
  roll: 16,
  name: 'Kaif',
  age: 22,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb02'),
  roll: 17,
  name: 'Muskan',
  age: 22,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb03'),
  roll: 18,
  name: 'Leher',
  age: 21,
  department: 'CSE'
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9caf8'),
  roll: 7,
  name: 'Adeel',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf9'),
  roll: 8,
  name: 'Nasib',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafa'),
  roll: 9,
  name: 'Arslan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafb'),
  roll: 10,
  name: 'Rehan',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafc'),
  roll: 11,
  name: 'Faheem',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafd'),
  roll: 12,
  name: 'Akhil',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId("6aae5e6585178ee815c9cb04"),
  roll: 19,
  name: 'Ashar',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId("6aae5e6585178ee815c9cb05"),
  roll: 20,
  name: 'Husnain',
  age: 21,
  department: 'ECA'
}
Type "it" for more
it
{
  _id: ObjectId("6aae5e6585178ee815c9cb06"),
  roll: 21,
  name: 'Jhon',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId("6aae5e6585178ee815c9cb07"),
  roll: 22,
  name: 'Shon',
  age: 22,
  department: 'CSE'
}
{
  _id: ObjectId("6aae5e6585178ee815c9cb08"),
  roll: 23,
  name: 'Nikhil',
  age: 22,
  department: 'ECA'
}
{
  _id: ObjectId("6aae5e6585178ee815c9cb09"),
  roll: 24,
  name: 'Nitish',
  age: 22,
  department: 'CSE'
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9cb0a'),
  roll: 25,
  name: 'Mayan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb0b'),
  roll: 26,
  name: 'Piyush',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb0c'),
  roll: 27,
  name: 'Ayan',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb0d'),
  roll: 28,
  name: 'Faizan',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb0e'),
  roll: 29,
  name: 'Palak',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb0f'),
  roll: 30,
  name: 'Hazel',
  age: 20,
  department: 'ME'
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9cb10'),
  roll: 31,
  name: 'Divyanshu',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb11'),
  roll: 32,
  name: 'Shivam',
  age: 20,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb12'),
  roll: 33,
  name: 'Ankush',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb13'),
  roll: 34,
  name: 'Aman',
  age: 22,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb14'),
  roll: 34,
  name: 'Abhinay',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb15'),
  roll: 35,
  name: 'Vinay',
  age: 20,
  department: 'CSE'
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9cb16'),
  roll: 36,
  name: 'Akash',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb17'),
  roll: 37,
  name: 'Anmol',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb18'),
  roll: 38,
  name: 'Kunal',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb19'),
  roll: 39,
  name: 'Anil',
  age: 22,
  department: 'CSE'
}
Type "it" for more
it
{
  _id: ObjectId('6aae5e6585178ee815c9cb1a'),
  roll: 40,
  name: 'Nitya',
  age: 21,
  department: 'CSE'
}
```

UPDATE OPERATION- The Update operation modifies existing documents. MongoDB provides methods like update One() to update a single document and update Many() to update multiple documents. The $set operator is commonly used to specify the fields to update.

```
db.CRUD.updateOne({roll:31},{$set:{department:"ECA"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9cb10'),
  roll: 31,
  name: 'Divyanshu',
  age: 21,
  department: 'ECA'
}
```

```
db.CRUD.updateMany({roll:22},{$set:{department:"ME"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db.CRUD.find()
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9cafb'),
  roll: 10,
  name: 'Rohan',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafc'),
  roll: 11,
  name: 'Faheem',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafd'),
  roll: 12,
  name: 'Akhil',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafe'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caff'),
  roll: 14,
  name: 'Mukesh',
  age: 20,
  department: 'ME'
}
```

DELETE OPERATION- The Delete operation removes documents from a collection. MongoDB offers deleteOne() to delete a single document and deleteMany() to remove multiple documents based on a filter.

```
> db.CRUD.deleteOne({roll:15})
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.CRUD.find()
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9caff'),
  roll: 14,
  name: 'Nukesh',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb01'),
  roll: 16,
  name: 'Kaif',
  age: 22,
  department: 'ME'
}
```

```
db.CRUD.deleteMany({department:"ME"})
{
  acknowledged: true,
  deletedCount: 16
}
db.CRUD.find()
```

```
db.CRUD.find()
{
  _id: ObjectId('6aae544e85178ee815c9caf2'),
  rollno: 1,
  name: 'Sam',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf3'),
  rollno: 2,
  name: 'Ram',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf4'),
  roll: 3,
  name: 'Adi',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf6'),
  roll: 5,
  name: 'Bee',
  age: 20,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf7'),
  roll: 6,
  name: 'Arif',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9caf8'),
  roll: 7,
  name: 'Adeel',
  age: 21,
  department: 'CSE'
}
```

```
{
  _id: ObjectId('6aae5e6585178ee815c9caf9'),
  roll: 8,
  name: 'Nasib',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafa'),
  roll: 9,
  name: 'Arslan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafc'),
  roll: 11,
  name: 'Faheem',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cafe'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb03'),
  roll: 18,
  name: 'Leher',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6585178ee815c9cb05'),
  roll: 20,
  name: 'Husnain',
  age: 21,
  department: 'ECA'
}
```

Using greater than in Mongo DB

```
> db.students.find({marks:{$gt:80}})
< {
    _id: ObjectId('6ab21b30555468477e03bd2e'),
    name: 'Amit',
    age: 21,
    marks: 85,
    department: 'CSE',
    skills: [
      'Python',
      'MongoDB',
      'Java'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd30'),
    name: 'Priya',
    age: 20,
    marks: 91,
    department: 'CSE',
    skills: [
      'Java',
      'MongoDB',
      'AI'
    ]
  }
```

Using less than in Mongo DB

```
> db.students.find({marks:{$lt:80}})
< {
    _id: ObjectId('6ab21b30555468477e03bd2f'),
    name: 'Rahul',
    age: 22,
    marks: 72,
    department: 'ECE',
    skills: [
      'Python',
      'SQL'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd31'),
    name: 'Neha',
    age: 23,
    marks: 65,
    department: 'ME',
    skills: [
      'AutoCAD',
      'Python'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd32'),
    name: 'Rohit',
    age: 21,
    marks: 78,
    department: 'CSE',
    skills: [
      'Python',
      'AI'
    ]
  }
```

Using equal to in Mongo DB

```
> db.students.find({marks:{$eq:78}})
< {
    _id: ObjectId('6ab21b30555468477e03bd32'),
    name: 'Rohit',
    age: 21,
    marks: 78,
    department: 'CSE',
    skills: [
      'Python',
      'AI'
    ]
  }
```

Using not equal to in Mongo DB

```
> db.students.find({marks:{$ne:85}})
< {
    _id: ObjectId('6ab21b30555468477e03bd2f'),
    name: 'Rahul',
    age: 22,
    marks: 72,
    department: 'ECE',
    skills: [
      'Python',
      'SQL'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd30'),
    name: 'Priya',
    age: 20,
    marks: 91,
    department: 'CSE',
    skills: [
      'Java',
      'MongoDB',
      'AI'
    ]
  }
```

Using greater and equal to in Mongo DB

```
> db.students.find({marks:{$gte:85}})
< {
    _id: ObjectId('6ab21b30555468477e03bd2e'),
    name: 'Amit',
    age: 21,
    marks: 85,
    department: 'CSE',
    skills: [
      'Python',
      'MongoDB',
      'Java'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd30'),
    name: 'Priya',
    age: 20,
    marks: 91,
    department: 'CSE',
    skills: [
      'Java',
      'MongoDB',
      'AI'
    ]
  }
```

Using Conditional Operation in Mongo DB

AND Operation:-

```
> db.students.find({$and:[{marks:{$gt:70}},{age:{$lt:23}}]})
< {
    _id: ObjectId('6ab21b30555468477e03bd2e'),
    name: 'Amit',
    age: 21,
    marks: 85,
    department: 'CSE',
    skills: [
      'Python',
      'MongoDB',
      'Java'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd2f'),
    name: 'Rahul',
    age: 22,
    marks: 72,
    department: 'ECE',
    skills: [
      'Python',
      'SQL'
    ]
  }
```

Using Conditional Operation in Mongo DB

OR Operation:-

```
> db.students.find({$or:[{department:"CSE"},{department:"ECA"}]})
< {
    _id: ObjectId('6ab21b30555468477e03bd2e'),
    name: 'Amit',
    age: 21,
    marks: 85,
    department: 'CSE',
    skills: [
      'Python',
      'MongoDB',
      'Java'
    ]
  }
  {
    _id: ObjectId('6ab21b30555468477e03bd30'),
    name: 'Priya',
    age: 20,
    marks: 91,
    department: 'CSE',
    skills: [
      'Java',
      'MongoDB',
      'AI'
    ]
  }
```